

A quantitative modeling of protein-DNA interaction and its application in regulatory genomics

Junguk Hur
School of Informatics

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Overview

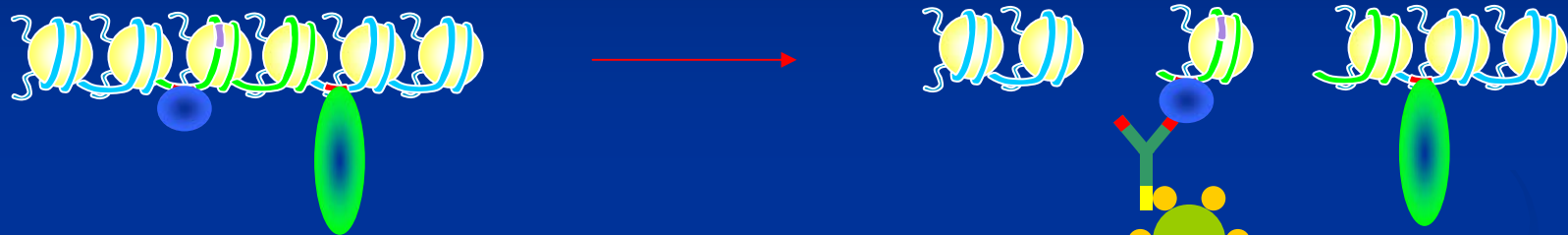
- Recent technologies for identification of protein-DNA interaction
- Quantitative modeling for transcriptional factor binding site motif discovery
- Our approaches

Recent Technologies

- Genome-wide location analysis
- *In vivo*: **ChIP-on-Chip** (Chromatin Immuno Precipitation on Microarray)
- *In vitro*: **PBM** (Protein Binding Microarray)

Chromatin Immuno Precipitation (ChIP)

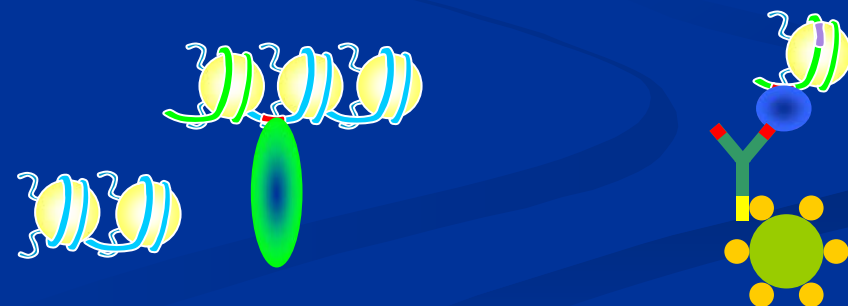
Sonication or
vortexing with glass-beads



Immunoprecipitation

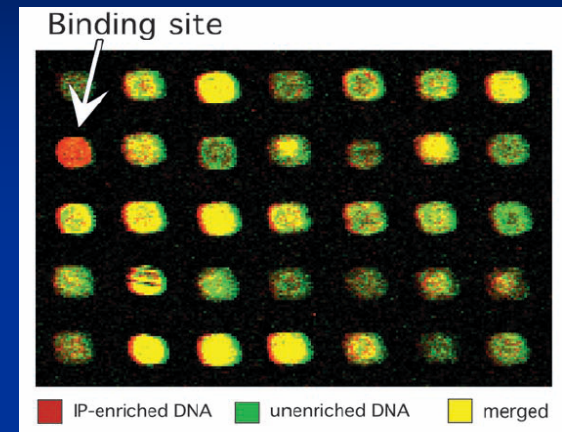
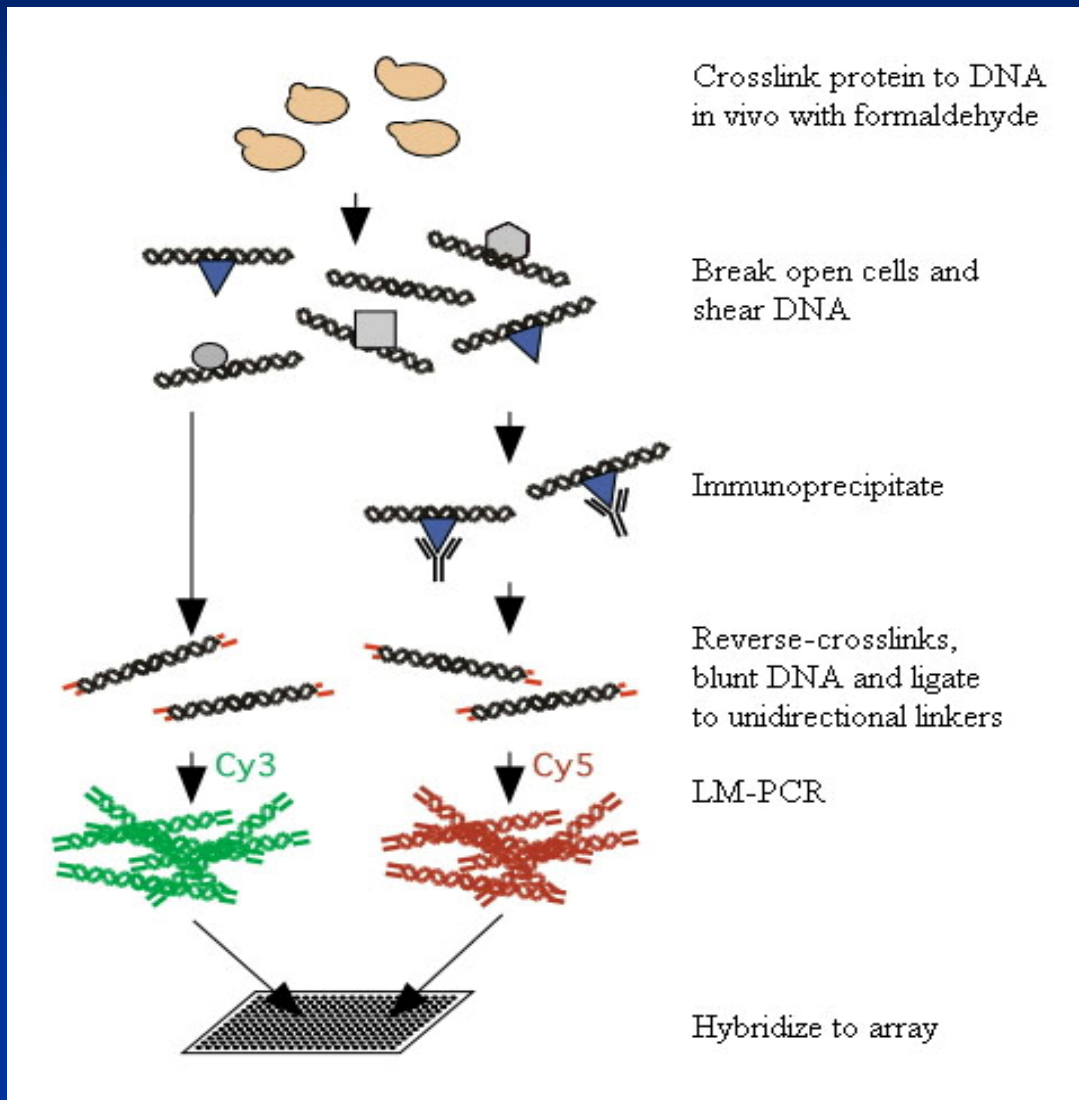
Supernatant

Pellet



- Using antibody of a protein of interest
- DNA bound to specific protein are enriched.

ChIP-on-Chip (Ren *et al.*)



	Galactose	Galactose
Name	p-value	average ratio
iYBR021W	0.000000121	10.3
iYLR080W	0.000000142	9.56
iYBR018C	0.00000038	8.36
iYDR008C	0.000000252	8.26
iYLR094C	0.000000302	7.94
iYBR019C	0.000000696	7.89
iYLR096W	0.00000247	5.4

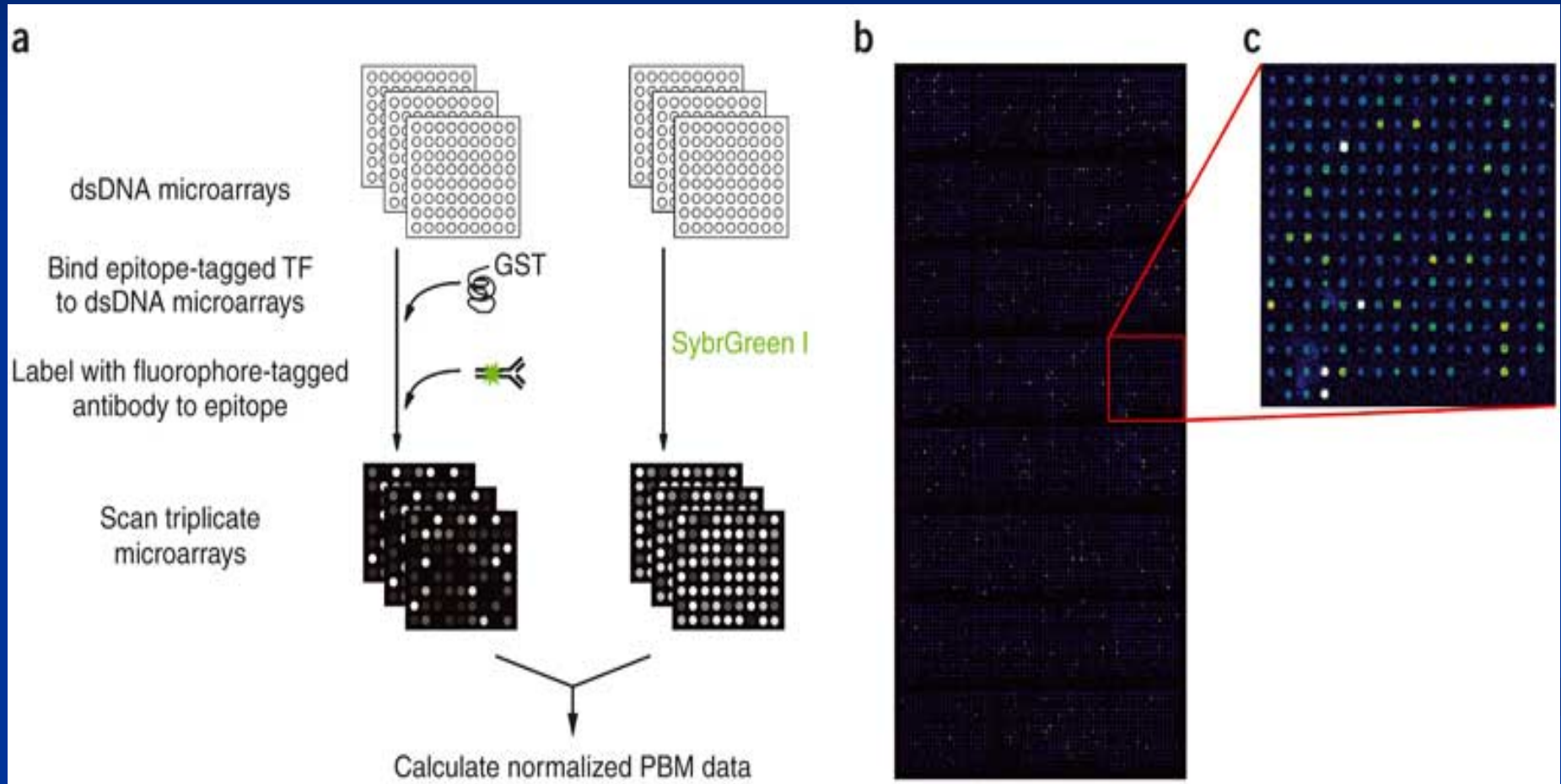
Array of **intergenic**
sequences from the whole
genome

Protein Binding Microarray (PBM)

(Bulyk *et al.*)

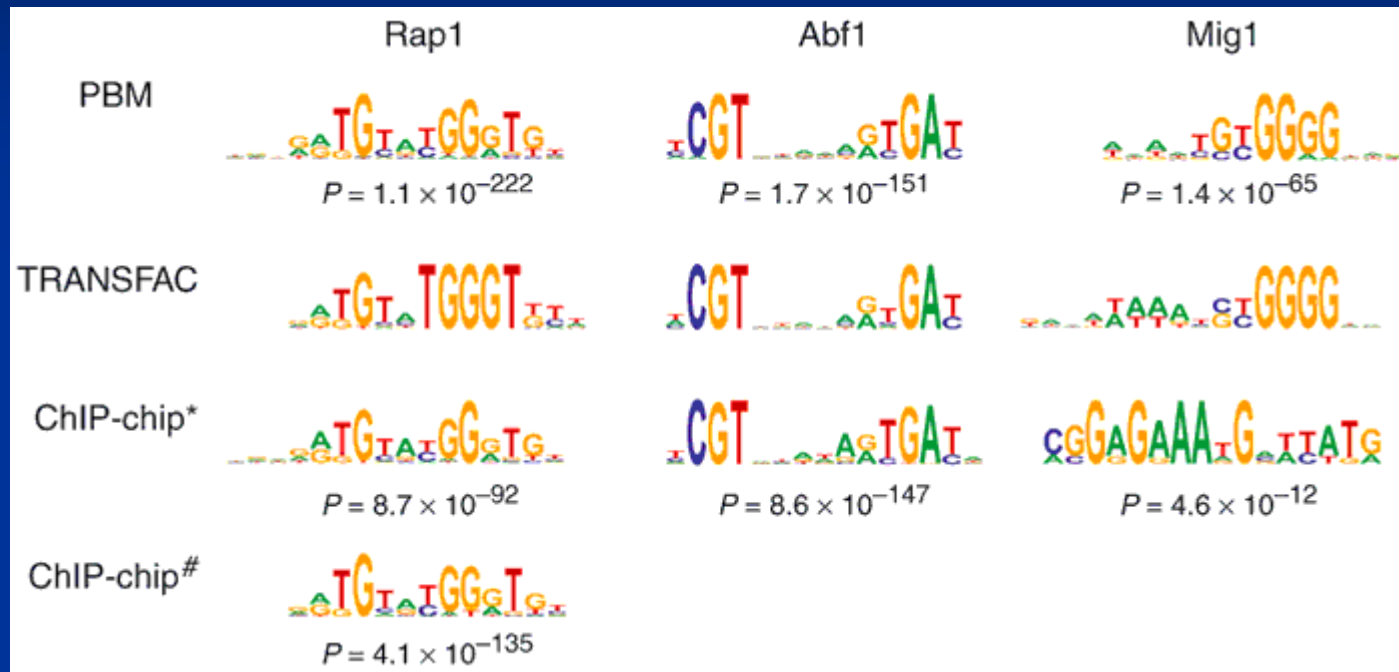
- *In vitro* assay
- DNA-binding protein of interest is expressed with an epitope tag, purified and then bound **directly to a double-strand DNA microarray**
- Can overcome the shortcomings of ChIP-on-Chip
 - Poor enrichment
 - No available antibody
 - Unknown culture condition or time points

Protein Binding Microarray



Whole-genome yeast intergenic microarray bound by Rap1

ChIP-on-Chip vs PBM



- Done by Mukherjee *et al.*
- Useful when ChIP-on-Chip does not result in enough enrichment
- * Lee *et al.* , # Lieb *et al.*

ChIP-on-Chip vs PBM

- *In vitro* assay - Not in the same state as it might be.
- Difference between ChIP-on-Chip and PBM may help identify whether there are substantial *in vivo* effects; chromatin structure or **cofactors** important in allowing or preventing sequence-specific binding.
- May contribute to deducing co-regulatory mechanisms

Motif Finding Algorithms

- MEME (Expectation Maximization)
- Gibbs (Gibbs Sampling)
- CONSENSUS (greedy multiple alignment)
- WINNOWER (Clique finding in graphs)
- SP-STAR (Sum of pairs scoring)
- MITRA (Mismatch trees to prune exhaustive search space)
- Same weight for each subsequences
- Not-quantitative

Quantitative Modeling

Traditional

CGGATGCATGCATGCCG	1
CGAAAGTGCATGACCCG	1
CCGGATCAGGTGACCCG	1
CGGAGATGGACTGACCG	1
AGGAAGATGCAGTGTCG	1
CAATGACGATGACATAT	1
CGGGAGATCAGGAGTTT	1

	Galactose	Galactose
Name	p-value	average ratio
iYBR021W	0.000000121	10.3
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iYBR018C	0.000000038	8.36
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iYBR019C	0.000000696	7.89
iYLR096W	0.00000247	5.4

Quantitative

CGGATGCATGCATGCCG	10.3
CGAAAGTGCATGACCCG	9.56
CCGGATCAGGTGACCCG	8.36
CGGAGATGGACTGACCG	8.26
AGGAAGATGCAGTGTCG	7.94
CAATGACGATGACATAT	7.89
CGGGAGACAGGTAGTTT	5.4

- More realistic reflecting actual binding affinity

Energy-Based Motif Finding (EBMF)

- Let e_i be the average binding energy between TF and sequence s_i , then $e_i = -\ln(K_e)$
$$K_e = [TF \cdot s_i] / [TF][s_i]$$
- Color intensity represents the value of K_e
- $4 \times l$ energy matrix M to represent the motif

Energy-Based Motif Finding (EBMF)

- Problem Definition
 - Minimize the prediction error

$$\sum_{i=1}^t (E_{total}(s_i, M) - e_i)^2$$

- Iteratively improve candidate matrix M

Energy-Based Motif Finding (EBMF)

■ Results by Chin *et al.*

	$E(L_E, B)$	EBMF		AlignACE		MEME	
		Find?	rank	Find?	rank	Find?	rank
$B = 7$	149475	yes	1	no	-	no	-
$B = 8$	0.000439	yes	1	no	-	yes	1
$B = 9$	7.70349×10^{-7}	yes	1	yes	1	yes	1

	Our Algorithm		AlignACE		MEME	
	Find?	rank	Find?	rank	Find?	rank
Using the top 100 sequences in the original data	yes	2	yes	1	yes	1
Using the top 100 sequences except sequences 2,3,4 and 6	yes	1	no	-	no	-
Using the top 100 sequences except sequences 1 to 6	yes	10	no	-	no	-
Using the top 100 sequences except sequences 1 to 8	yes	5	no	-	no	-

■ Out-performed MEME & AlignACE

Our Approaches

- Build better model representing the local and non-local correlation between nucleotides
 - Based on the EBMF algorithm
 - Utilizing quantitative measure for DNA-protein interaction
 - Potentially more accurate than the Positional Weight Matrices (PWMs)
- Deduce correlations between TFs

Our Approaches

- Find highly correlated positions

	1	2	3	~	-1	
A	.5	.2	.1	~		.4
C	.2	.2	.1	~		.2
G	.2	.2	.1	~		.1
T	.1	.4	.7	~		.3

4 x 7



	1 & 2
AA	0.10
AC	0.10
AG	0.10
AT	0.40
CA	0.05
~	
TT	0.08

16 x 1

+

	3	~	-1	
A	.1	~		.4
C	.1	~		.2
G	.1	~		.1
T	.7	~		.3

4 x (7-1)

Iteratively improve candidate matrix M

Our Approaches

- Enumerate searching spaces for all informative combination of positions
- Solving linear equations
 - *QR Decomposition - Fast but cannot solve singular matrix*
 - *Least Square Method - Slow but can solve singular matrix*

Future Developments

- Whole-genome analysis
- Differential motif analysis btw different experiments
 - Derive the differential binding affinities of TFs
 - *In vivo* vs *In vitro*
 - Different environments
 - Extract co-regulatory mechanisms or interactions btw TFs

Thank you

Any question and suggestion ?